

AJ Quisno

DSC140S

Dr. Butler

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Ohio vs Everybody: A Statistical Analysis of One of the Greatest CFB Programs of All Time

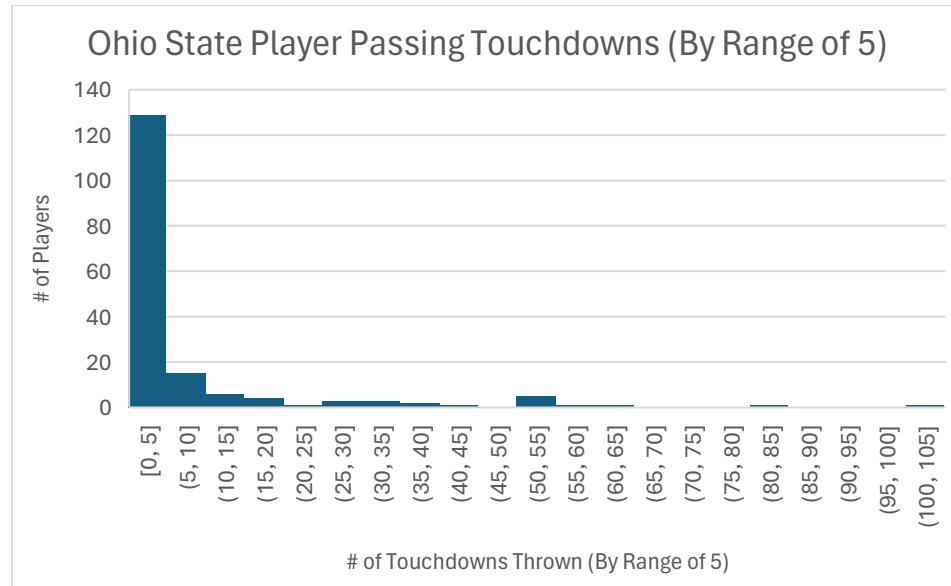
Since the game's inception, The Ohio State University Buckeyes have been one of the most dominating forces throughout college football (Nagel). Given the Buckeyes being a dominating force in the college football world, researching and analyzing the statistics of this great team is essential to understanding why they have been so successful, especially when it comes to passing stats and quarterback play. Three important questions to analyze are how many players to attempt a pass in Ohio State History have thrown at least 1 touchdown, can the number of touchdowns and completions a player has thrown at Ohio State be used to predict the number of years a player has played at Ohio State, through machine learning, and do the years a player has played at Ohio State have any impact on the player's passing rating? These questions could possibly provide valuable insight not only into team statistics but who and what players have been the most impactful in the team's history when it comes to attacking through the air offensively (College Football Reference). This data was chosen for analysis due to a long time fanhood within the community and a passion for sports data especially data pertaining to American football. This data set specifically contains information on all the players to ever attempt a pass in the history of the school's program, totaling 173 players. The columns of the data set contain important passing stats like completions, passer rating, interceptions, player name, etc. Below is what the data set looked like during analysis.

Figure 1: Ohio State Passing Stats Data Set

Rk	Player	Yrs	From	To	G	Cmpg	Att	Cmpg%	Yds	TD	TD%	Int	Int%	Y/A	AY/A	Y/C	WG	Rate	Player-additional
1	J.T. Barnett	4	2014	2017	50	799	1211	65.5	9434	104	8.6	33	2.5	7.8	8.39	12.3	188.7	152.3	j-t-barnett-1
2	C.L. Stroud	3	2020	2022	25	575	830	69.3	8123	65	10.2	17	1.4	9.8	11.18	14.1	324.9	162.4	c-l-stroud-1
3	Bobby Hoy	4	1992	1995	42	463	762	59.2	6791	54	6.9	33	4.2	8.6	8.12	14.6	169.7	146.1	bobby-hoying-1
4	Art Schlicht	4	1978	1981	44	439	849	51.7	6384	44	5.2	42	4.9	7.8	6.07	15	149.5	124.1	art-schlichter-1
5	Terrill Pryor	3	2008	2010	39	477	783	60.9	6177	57	7.3	26	3.3	7.9	7.85	12.9	158.4	144.6	terrell-pryor-1
6	Greg Frey	4	1987	1990	44	417	777	53.7	5074	36	4.6	35	4.5	7.7	6.59	14.3	135.8	124.5	greg-frey-1
7	Joe Germaine	3	1996	1998	34	399	660	60.5	5844	52	7.9	18	2.7	8.9	9.2	14.6	171.9	155.4	joe-germaine-1
8	Troy Smith	4	2003	2006	32	420	670	62.7	5720	54	8.1	13	1.9	8.5	9.28	13.6	178.8	157.1	troy-smith-1
9	Steve Belli	4	1998	2001	43	351	699	50.2	5401	33	4.7	27	3.9	7.7	6.93	15.4	125.5	123	steve-bellisari-1
10	Dwayne H	2	2017	2018	22	413	590	70	5396	54	9.2	9	1.5	9.1	10.29	13.1	245.3	174	dwayne-haskins-1
11	Justin Fields	2	2019	2020	22	396	579	68.4	5375	63	10.9	9	1.6	9.3	10.76	13.6	244.2	179.1	justin-fields-2
12	Braxton M	5	2011	2015	49	396	667	59.4	5295	52	7.8	17	2.5	7.9	6.35	13.4	193.1	146.7	braxton-miller-1
13	Mike Tomczak	4	1981	1984	44	325	588	55.4	4921	30	5.1	32	5.4	8.4	6.94	15.1	111.8	131.7	mike-tomczak-1
14	Jim Karsatos	4	1983	1986	41	330	573	57.6	4698	36	6.3	19	3.3	8.2	7.96	14.2	114.5	140.6	jim-karsatos-1
15	Craig Kneer	4	2000	2003	32	326	574	56.8	4473	28	4.9	29	3.5	7.8	7.2	13.7	139.8	131.4	craig-kneer-1
16	Will Howard	1	2024	2024	15	292	402	72.6	3779	33	8.2	10	2.5	9.4	9.92	12.9	201.9	173.7	will-howard-2
17	Kyle McCoy	3	2021	2023	24	270	406	66.5	3776	27	6.7	8	2	9.3	9.74	14	157.3	162.6	kyle-mccord-1
18	Todd Boschee	4	2005	2008	28	255	402	63.4	5085	31	7.7	16	4	7.7	7.43	12.1	110.2	145.4	todd-boschee-1
19	Stanley Jackson	4	1994	1997	41	182	329	55.3	2567	21	8.4	8	2.4	7.8	7.98	14.1	62.5	157.1	stanley-jackson-1
20	Don Unnewehr	3	1963	1965	27	220	468	47	2518	12	2.6	26	5.8	5.4	3.39	11.4	93.3	88.6	don-unnewehr-1
21	Rex Kern	3	1968	1970	20	175	336	52.1	2303	17	5.1	23	6.8	6.9	4.79	13.2	92.1	112.7	rex-kern-1
22	Kirk Herbstreit	4	1989	1992	38	183	317	57.7	2263	5	1.6	11	3.5	7.1	5.89	12.4	59.5	116	kirk-herbstreit-1
23	Tom Tupai	4	1984	1987	34	168	296	56.8	2229	15	5.1	11	3.7	7.5	6.87	13.3	65.6	129.3	tom-tupai-1

The first question asked in this analysis was how many players that have attempted a pass in Ohio State history thrown at least one touchdown? This is a valid question because it could potentially provide insight into the success the Ohio State passing game has had. To try and figure this out, the original plan was to make a bar chart of the count of the players who have thrown a specific number of touchdowns. Excel defaulted to a range of five touchdowns each.

Figure 2.1: Bar Chart of Ohio State Player Passing Touchdowns (By Range of 5)



Obviously, this graph appears to be very hard to determine exactly how many players have thrown touchdowns within a range of five. Additionally, there is no way to correctly interpret the number of players who have specifically thrown at least one touchdown due to there being a range of five to begin with. To combat this and get a more accurate analysis, it was determined that a pivot table would need to be created. It was determined that the Pivot Table would contain the count of the players as well as filtration via the number of touchdowns a player has thrown. The pivot table was created and filtered to contain the number of players who have thrown equal to or greater than one touchdown.

Figure 2.2: Pivot Table of Ohio State Players who have thrown at least 1 Passing Touchdown

TD	(Multiple Items)
Count of Player	91

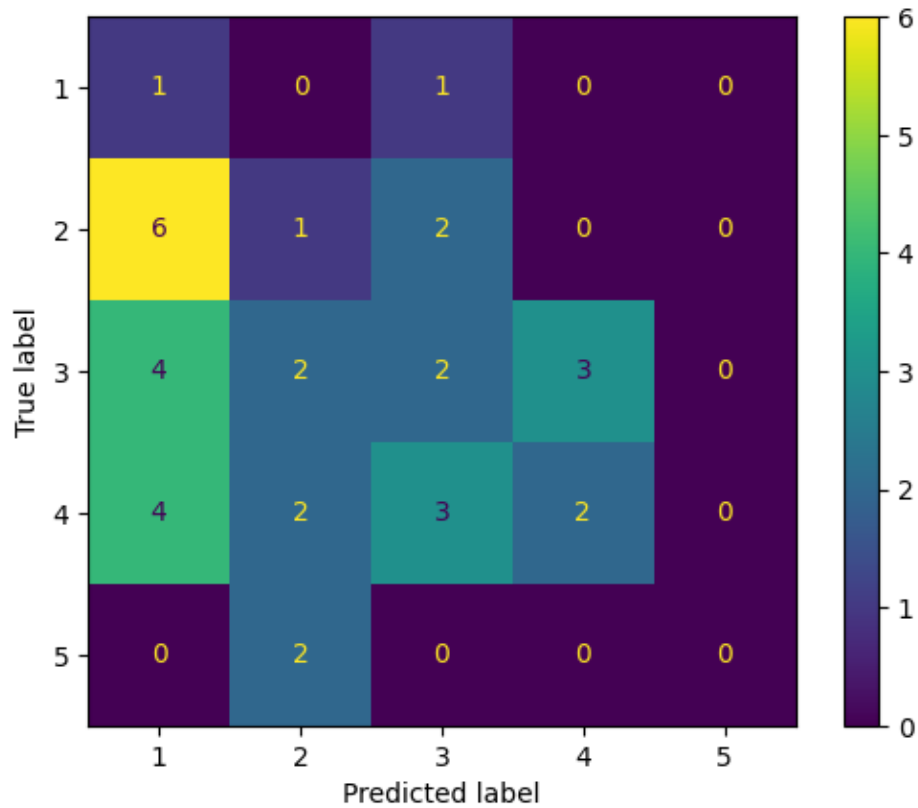
Thanks to the pivot table, it was accurately determined that the number of players in Ohio State history to throw at least one passing touchdown was 91 out of 173 players. To conclude, a majority of players that have attempted a pass in Ohio State history have thrown a touchdown.

Note: Data includes all players to attempt a pass, not just quarterbacks (College Football Reference).

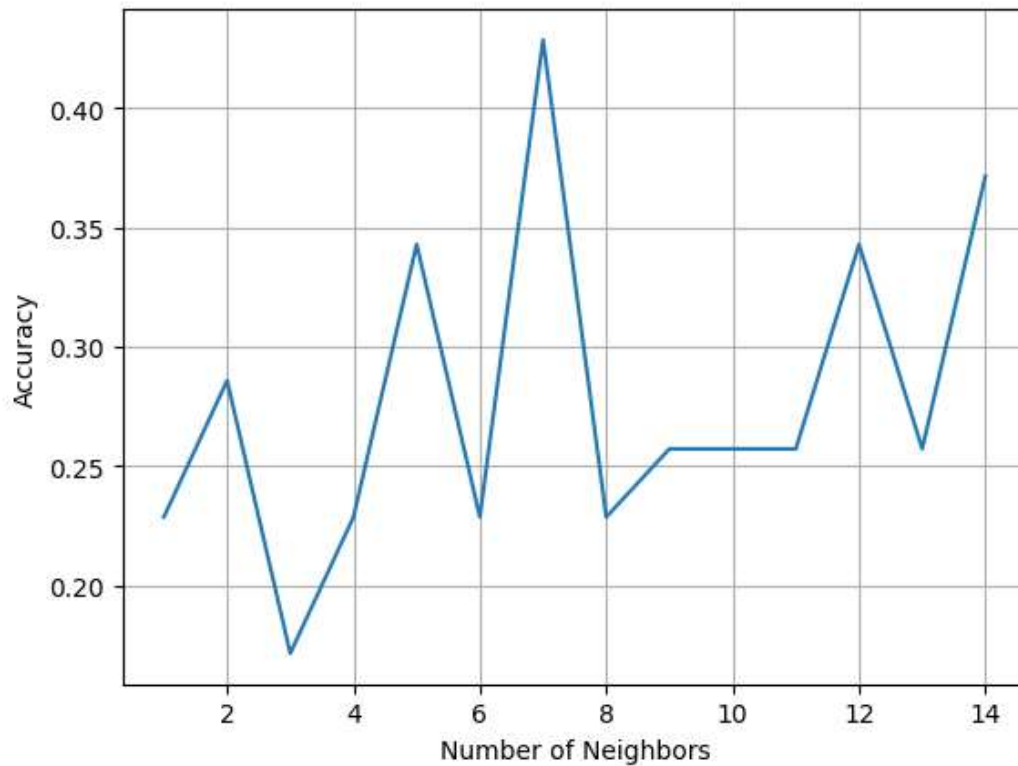
The second question asked during this analysis was can the number of touchdowns and completions a player has thrown at Ohio State be used to predict the number of years a player has played at Ohio State? This question is important because it could be used to help predict the statistics of highly coveted incoming freshmen and provide insight into what is expected of them in terms of performance. To complete this, supervised machine learning was conducted in Python using the K-Nearest Neighbors Algorithm. The data frame was first divided into their respective train and test sets, the y-data containing the years played, and the x-data containing the touchdowns and completions thrown. It was decided to start with three neighbors for no particular reason. The accuracy score for the three-neighbor prediction test was 17.14%, which is extremely poor. A confusion matrix was displayed containing the specifics of what the machine was able to correctly predict and what the machine predicted incorrectly.

Figure 3.1: 3-Neighbor Years Played Confusion Matrix

```
[14]: <sklearn.metrics._plot.confusion_matrix.ConfusionMatrixDisplay at 0x19b77912660>
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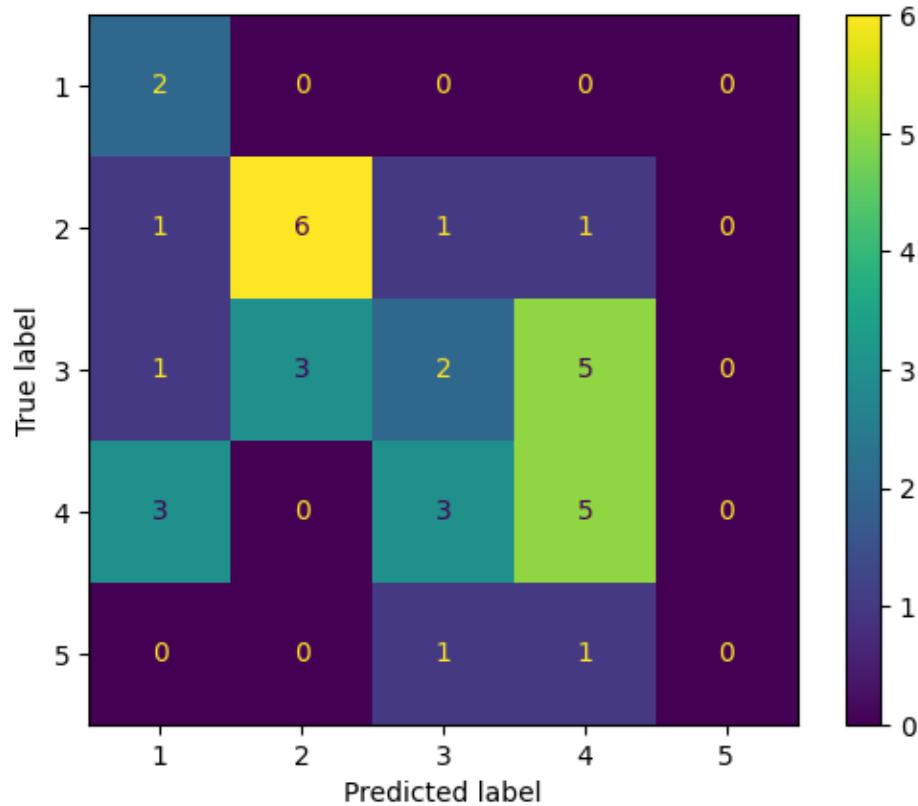
As shown, the machine rarely made any correct predictions. To help increase the accuracy of the machine, hyperparameter tuning was used in order to determine the most optimal number of neighbors for the highest accuracy score. Below is a line graph containing the number of neighbors corresponding with their potential accuracy score.

Figure 3.2: Line Graph of Optimal Neighbors w/ Potential Accuracy Scores

According to this graph, the optimal number of neighbors is seven. Predictions were remade by the machine with seven neighbors instead of three, which produced an accuracy score of about 42.86%. Although a vast improvement, this accuracy is still considered to be very poor. A confusion matrix was displayed containing the specifics of what the machine was able to correctly predict and what the machine predicted incorrectly.

Figure 3.3: 7-Neighbor Years Played Confusion Matrix

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[20]: <sklearn.metrics._plot.confusion_matrix.ConfusionMatrixDisplay at 0x19b7d3779d0>
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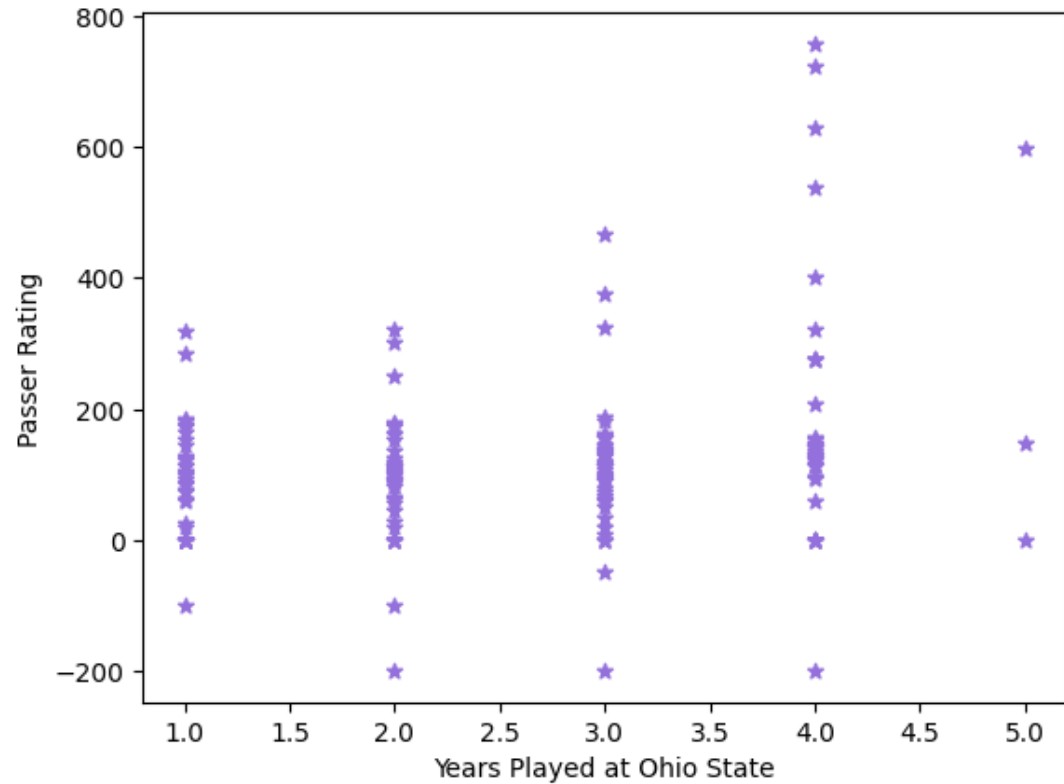


To summarize, it is not possible to use the number of touchdowns and completions a player has thrown to predict the number of years a player has played at Ohio State using the K-Nearest Neighbors algorithm.

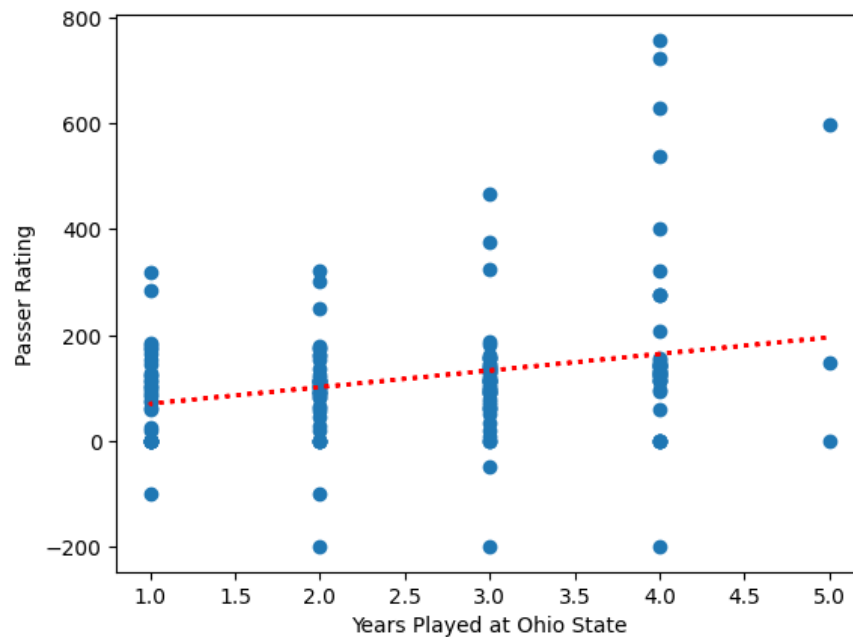
The third question asked in this analysis was do the years a player has played at Ohio State have any impact on the player's passer rating? This question is important because it provides insight into whether or not a player can improve and grow if they stay at Ohio State for all of their four to five years as a player. To solve this, a scatter plot was made within the Matplotlib Python library, containing the years played on the x-axis, and the passer rating on the y-axis.

Figure 4.1: Scatter Plot of Years Played over Passer Rating

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[33]: Text(0, 0.5, 'Passer Rating')
```



A lot can be inferred from this scatter plot, such as the fact that there appears to be a positive correlation between the years played and the passer rating. After the creation of this scatter plot, a Pearson's correlation test was conducted between the two data columns. It was calculated that there was a slight positive correlation between the years played and the passer rating, a score of about 0.27. To prove this graphically, the scatter plot was remade, and a linear model was created to help determine and plot a line of best fit across the scatter plot.

Figure 4.2: Scatter Plot of Years Played over Passer Rating w/Line of Best Fit

To conclude, there is a positive impact and correlation between the years a player has played at Ohio State and their passer rating.

Ohio State is one of the few college football programs that has continued to be dominant throughout its entire existence and is widely regarded as one of the best. This analysis serves as nothing more than a proving point to Ohio State's excellence in the passing game throughout the program's tenure. The three questions asked provided valuable insight into the statistics and history that help prove the high quality of Ohio State's passers.

Works Cited

College Football Reference. "Ohio State Buckeyes Passing Stats | College Football at Sports-Reference.com." *College Football at Sports-Reference.com*, 2024, www.sports-reference.com/cfb/schools/ohio-state/passing.html. Accessed 12 Apr. 2025.

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